PLSC 504 – Fall 2026

Regression Models for Binary and Nominal Responses

August 31, 2026

Latent:

$$Y_i^* = \mathbf{X}_i\boldsymbol{\beta} + u_i$$

Observed:

$$\begin{aligned} Y_i &= 0 \text{ if } Y_i^* < 0 \\ Y_i &= 1 \text{ if } Y_i^* \geq 0 \end{aligned}$$

So:

$$\begin{aligned} \Pr(Y_i = 1) &= \Pr(Y_i^* \geq 0) \\ &= \Pr(\mathbf{X}_i\boldsymbol{\beta} + u_i \geq 0) \\ &= \Pr(u_i \geq -\mathbf{X}_i\beta) \\ &= \Pr(u_i \leq \mathbf{X}_i\beta) \\ &= \int_{-\infty}^{\mathbf{X}_i\boldsymbol{\beta}} f(u)du \end{aligned}$$

"Standard logistic" PDF:

$$\Pr(u) \equiv \lambda(u) = \frac{\exp(u)}{[1 + \exp(u)]^2}$$

CDF:

$$\begin{aligned} \Lambda(u) &= \int \lambda(u) du \\ &= \frac{\exp(u)}{1 + \exp(u)} \\ &= \frac{1}{1 + \exp(-u)} \end{aligned}$$

Logistic → "Logit"

$$\begin{aligned}
\Pr(Y_i = 1) &= \Pr(Y_i^* > 0) \\
&= \Pr(u_i \leq \mathbf{X}_i\boldsymbol{\beta}) \\
&= \Lambda(\mathbf{X}_i\boldsymbol{\beta}) \\
&= \frac{\exp(\mathbf{X}_i\boldsymbol{\beta})}{1 + \exp(\mathbf{X}_i\boldsymbol{\beta})} \\
\text{(equivalently)} &= \frac{1}{1 + \exp(-\mathbf{X}_i\boldsymbol{\beta})}
\end{aligned}$$

$$L = \prod_{i=1}^{N} \left(\frac{\exp(\mathbf{X}_i\boldsymbol{\beta})}{1 + \exp(\mathbf{X}_i\boldsymbol{\beta})} \right)^{Y_i} \left[1 - \left(\frac{\exp(\mathbf{X}_i\boldsymbol{\beta})}{1 + \exp(\mathbf{X}_i\boldsymbol{\beta})} \right) \right]^{1-Y_i}$$

$$\ln L = \sum_{i=1}^{N} Y_i \ln \left(\frac{\exp(\mathbf{X}_i\boldsymbol{\beta})}{1 + \exp(\mathbf{X}_i\boldsymbol{\beta})} \right) + (1 - Y_i) \ln \left[1 - \left(\frac{\exp(\mathbf{X}_i\boldsymbol{\beta})}{1 + \exp(\mathbf{X}_i\boldsymbol{\beta})} \right) \right]$$

Normal → "Probit"

$$\begin{aligned}\Pr(Y_i = 1) &= \Phi(\mathbf{X}_i\boldsymbol{\beta}) \\ &= \int_{-\infty}^{\mathbf{X}_i\boldsymbol{\beta}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(\mathbf{X}_i\boldsymbol{\beta})^2}{2}\right) d\mathbf{X}_i\boldsymbol{\beta}\end{aligned}$$

$$L = \prod_{i=1}^{N} \left[\Phi(\mathbf{X}_i\boldsymbol{\beta})\right]^{Y_i} \left[1 - \Phi(\mathbf{X}_i\boldsymbol{\beta})\right]^{(1-Y_i)}$$

$$\ln L = \sum_{i=1}^{N} Y_i \ln\Phi(\mathbf{X}_i\boldsymbol{\beta}) + (1 - Y_i) \ln[1 - \Phi(\mathbf{X}_i\boldsymbol{\beta})]$$

Logit and Probit, Explained

Things we talked about in PLSC 503:

- Odds ratios and the random utility model
- Model estimation and interpretation
- Marginal effects, predictions, etc.
- Assessing model fit
- A couple variants (e.g., c-log-log)

Things:

- Models for dealing with “separation”
- Models for *rare events*

Common Focus: Shortage of information on Y

Separation

"Separation" = "perfect prediction" = "monotone likelihood"

Intuition: House votes on the PPACA (3/21/2010):

```
      Dems
Yeas    0   1
   0  178  34
   1    0 219
```

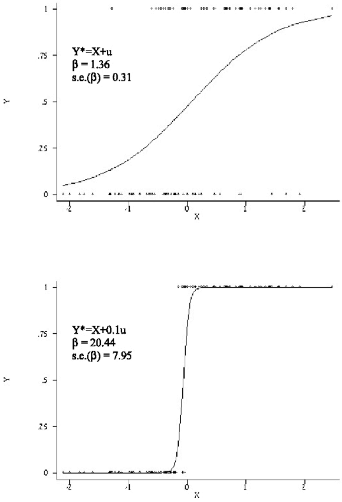
$\Pr(Y = 1 | X = 0) = ?$

"Separation" means that:

- $\hat{\beta}_X = \pm\infty$
- $\widehat{\text{s.e.}}_\beta = \infty$
- $\left.\frac{\partial^2 \ln L}{\partial X^2}\right|_{\hat{\beta}} = 0$ (monotone likelihood)

Separation Illustrated

Figure 1: Actual and Predicted Values, Simulated Logistic Regressions



Separation: What Happens

```
> set.seed(7222009)
> Z<-rnorm(500)
> W<-rnorm(500)
> Y<-rbinom(500,size=1,prob=plogis((0.2+0.5*W-0.5*Z)))
> X<-rbinom(500,1,(pnorm(Z)))
> X<-ifelse(Y==0,0,X)  # Induce separation of Y on X

> summary(glm(Y~W+Z+X,family="binomial"))

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   -0.638      0.133   -4.81  1.5e-06 ***
W              0.653      0.140    4.67  3.0e-06 ***
Z             -1.134      0.146   -7.76  8.3e-15 ***
X             20.915    861.458    0.02     0.98
---
Number of Fisher Scoring iterations: 18


# Change the maximum # of iterations / convergence tolerance:

> summary(glm(Y~W+Z+X,family="binomial",maxit=100,epsilon=1e-16))

Coefficients:
               Estimate  Std. Error z value Pr(>|z|)
(Intercept)      -0.638       0.133   -4.81  1.5e-06 ***
W                 0.653       0.140    4.67  3.0e-06 ***
Z                -1.134       0.146   -7.76  8.3e-15 ***
X                34.915 5978532.779    0.00        1
---
Number of Fisher Scoring iterations: 32

Warning message:
glm.fit: fitted probabilities numerically 0 or 1 occurred
```

One Solution: Exact Logistic Regression

Exact logistic regression (ELR):

- Cox (1970, Ch. 4); Hirji et al. (1987 *JASA*); Mehta & Patel (1995 *Stat. Med.*); Forster et al. (2003 *Stat. & Comp.*); Zamar and Graham (2007 *J. Stat. Soft.*).

- Conditions on permutations of covariate patterns

- $\longrightarrow$ Always has finite solutions for $\hat{\beta}$

- Implementation:
    - `elrm` in R ; `exlogistic` in Stata
    - Fitted via MCMC; see Forster et al. for details
    - In practice, there are often computational issues...

Firth's (1993) Correction

Firth proposed:

$$L(\boldsymbol{\beta}|Y)^{*} = L(\boldsymbol{\beta}|Y)\,|\mathbf{I}(\boldsymbol{\beta})|^{\frac{1}{2}}$$

$$\ln L(\boldsymbol{\beta}|Y)^{*} = \ln L(\boldsymbol{\beta}|Y) + 0.5 \ln |\mathbf{I}(\boldsymbol{\beta})|$$

"Penalized likelihood":

- Is consistent
- Eliminates small-sample bias
- Exist given separation
- To Bayesians, it's "Jeffreys' prior":

$$P(\theta) = \sqrt{\det\left[I(\theta)\right]}$$

- "Profile" (= "concentrated") likelihood
- $\hat{\beta}$ can be asymmetrical...
- $\rightarrow$ can affect "normal" inference...
- Plotting the profile likelihood and calculating alternative C.I.s is recommended

Two directions:

- R
  - `elrm` (exact logistic regression via MCMC)
  - `brlr` ("bias-reduced logistic regression")
  - `logistf` ("Firth's logistic regression")

- Stata
  - `exlogistic` (exact logistic regression)
  - `firthlogit` (Firth corrected logit)

Some data, and a silly question:

- CBS/NYT Poll, April 1997
- Standard political/demographics, plus
- "Do you consider your pet to be a member of your family, or not?"
- Yes = 84.4%, No = 15.6%

Data:

```
> summary(Pets)

   petfamily         female                married              partyid              education  
 Min.   :0.000   Min.   :0.000   Married       :442   Democrat   :225   < HS          : 71  
 1st Qu.:1.000   1st Qu.:0.000   Widowed       : 46   Independent:214   HS diploma    :244  
 Median :1.000   Median :1.000   Divorced/Sep:118   GOP        :229   Some college:184  
 Mean   :0.844   Mean   :0.556   NBM           :118   NA's       : 58   College Grad:131  
 3rd Qu.:1.000   3rd Qu.:1.000   NA's          :  2                     Post-Grad     : 96  
 Max.   :1.000   Max.   :1.000
```

Pets as Family: Basic Model

```
> Pets.1<-glm(petfamily~female+as.factor(married)+as.factor(partyid)
+             +as.factor(education),data=Pets,family=binomial)
> summary(Pets.1)

Coefficients:
                                Estimate Std. Error z value Pr(>|z|)
(Intercept)                       2.0133     0.5388    3.74  0.00019 ***
femaleMale                       -0.6959     0.2142   -3.25  0.00116 **
as.factor(married)Married        -0.0657     0.2911   -0.23  0.82147
as.factor(married)NBM             0.4599     0.3957    1.16  0.24504
as.factor(married)Widowed        -0.1568     0.4921   -0.32  0.75007
as.factor(partyid)Democrat       -0.1241     0.4286   -0.29  0.77213
as.factor(partyid)GOP            -0.0350     0.4321   -0.08  0.93537
as.factor(partyid)Independent    -0.1521     0.4299   -0.35  0.72338
as.factor(education)College Grad  0.2511     0.4121    0.61  0.54228
as.factor(education)HS diploma    0.0595     0.3685    0.16  0.87182
as.factor(education)Post-Grad     0.1946     0.4331    0.45  0.65321
as.factor(education)Some college  0.0587     0.3867    0.15  0.87928
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1   1

    Null deviance: 627.14  on 723  degrees of freedom
Residual deviance: 612.76  on 712  degrees of freedom
AIC: 636.8

Number of Fisher Scoring iterations: 4
```


THE SIMPSONS
CRAZY CAT LADY

Pets as Family: More Complicated Model

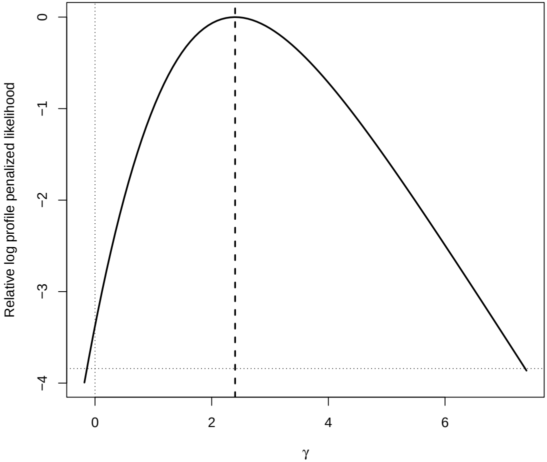
```
> Pets.2<-glm(petfamily~female+as.factor(married)*female+as.factor(partyid)+
+               as.factor(education),data=Pets,family=binomial)

> summary(Pets.2)

Coefficients:
                                     Estimate Std. Error z value Pr(>|z|)
(Intercept)                            2.2971     0.6166    3.73   0.0002 ***
femaleMale                            -1.1833     0.5305   -2.23   0.0257 *
as.factor(married)Married             -0.3218     0.4470   -0.72   0.4716
as.factor(married)NBM                  0.1854     0.6140    0.30   0.7628
as.factor(married)Widowed             -0.7415     0.5780   -1.28   0.1995
as.factor(partyid)Democrat            -0.1575     0.4297   -0.37   0.7140
as.factor(partyid)GOP                 -0.0445     0.4334   -0.10   0.9182
as.factor(partyid)Independent         -0.1757     0.4312   -0.41   0.6837
as.factor(education)College Grad       0.2332     0.4137    0.56   0.5730
as.factor(education)HS diploma         0.0558     0.3703    0.15   0.8801
as.factor(education)Post-Grad          0.2171     0.4342    0.50   0.6171
as.factor(education)Some college       0.0358     0.3890    0.09   0.9266
femaleMale:as.factor(married)Married   0.4853     0.5908    0.82   0.4114
femaleMale:as.factor(married)NBM       0.5260     0.8051    0.65   0.5136
femaleMale:as.factor(married)Widowed  15.2516   549.3719    0.03   0.9779
---

    Null deviance: 627.14  on 723  degrees of freedom
Residual deviance: 607.42  on 709  degrees of freedom
AIC: 637.4

Number of Fisher Scoring iterations: 14
```

```
> with(Pets, xtabs(~petfamily+as.factor(married)+female))

, , female = Female

         as.factor(married)
petfamily Divorced/Sep Married NBM Widowed
        0            7      28   5       7
        1           67     199  58      32

, , female = Male

         as.factor(married)
petfamily Divorced/Sep Married NBM Widowed
        0           11      47   8       0
        1           33     168  47       7
```

Pets as Family: Firth Model

```
> Pets.Firth<-logistf(petfamily~female+
+                       as.factor(married)*female+as.factor(partyid)+
+                       as.factor(education),data=Pets,flic=TRUE)

> summary(Pets.Firth)
logistf(formula = petfamily ~ female + as.factor(married) * female +
    as.factor(partyid) + as.factor(education), data = Pets, flic = TRUE)

Model fitted by Penalized ML
Coefficients:
                                         coef se(coef) lower 0.95 upper 0.95    Chisq         p method
(Intercept)                           2.21205    0.608      1.019      3.405 16.17636 0.0000577      1
femaleMale                           -1.13866    0.526     -2.187     -0.145  5.04186 0.0247420      2
as.factor(married)Married            -0.27387    0.442     -1.192      0.531  0.41518 0.5193531      2
as.factor(married)NBM                 0.15888    0.601     -0.991      1.367  0.07322 0.7867048      2
as.factor(married)Widowed            -0.72627    0.572     -1.839      0.384  1.67233 0.1959467      2
as.factor(partyid)Democrat           -0.11818    0.425     -0.992      0.661  0.08159 0.7751592      2
as.factor(partyid)GOP                -0.00776    0.429     -0.888      0.780  0.00034 0.9852893      2
as.factor(partyid)Independent        -0.13643    0.427     -1.013      0.646  0.10813 0.7422784      2
as.factor(education)College Grad      0.23904    0.411     -0.574      1.024  0.34480 0.5570689      2
as.factor(education)HS diploma        0.07531    0.368     -0.667      0.763  0.04289 0.8359331      2
as.factor(education)Post-Grad         0.21837    0.433     -0.627      1.050  0.26307 0.6080189      2
as.factor(education)Some college      0.05240    0.387     -0.721      0.781  0.01888 0.8906980      2
femaleMale:as.factor(married)Married  0.45582    0.588     -0.661      1.613  0.63550 0.4253467      2
femaleMale:as.factor(married)NBM      0.52329    0.794     -1.023      2.050  0.45133 0.5017022      2
femaleMale:as.factor(married)Widowed  2.40167    1.722     -0.139      7.374  3.37453 0.0662116      2

Method: 1-Wald, 2-Profile penalized log-likelihood, 3-None

Likelihood ratio test=17.3 on 14 df, p=0.242, n=724
Wald test = 263 on 14 df, p = 0
```

Profile Likelihood Plot



Note: Plot shows estimated profile likelihood for different values of the parameter estimate for the interaction term `femaleMale:as.factor(married)Widowed`. Horizontal dotted line is the likelihood associated with $P \leq 0.05$. Vertical dashed line is $\hat{\gamma}$; vertical dotted line indicates $\hat{\gamma} = 0$.

Wrap-Up

To summarize:

- Separation is an *estimation* problem...
- Separation $\nrightarrow$ dropping covariates!
- Firth's approach $>$ ELR
- Can also be applied to other sparse-data situations:
  - "Fixed effects" logit models (Cook et al. 2020)
  - Multinomial logit (Cook et al. 2018)
  - Survival models (Anderson et al. 2020)

Finally: Read this twitter thread before it's gone.

“Rare” Events

If events (“1s”) are rare, we can...

- Collect lots of “0s” for a few “1s”
- $\rightarrow$ Classification bias...

Example: Suppose that:

$$\Pr(Y_i) = \Lambda(0 + 1X_i)$$

then:

$$E(\hat{\beta}_0 - \beta_0) \approx \frac{\bar{\pi} - 0.5}{N\bar{\pi}(1 - \bar{\pi})}$$

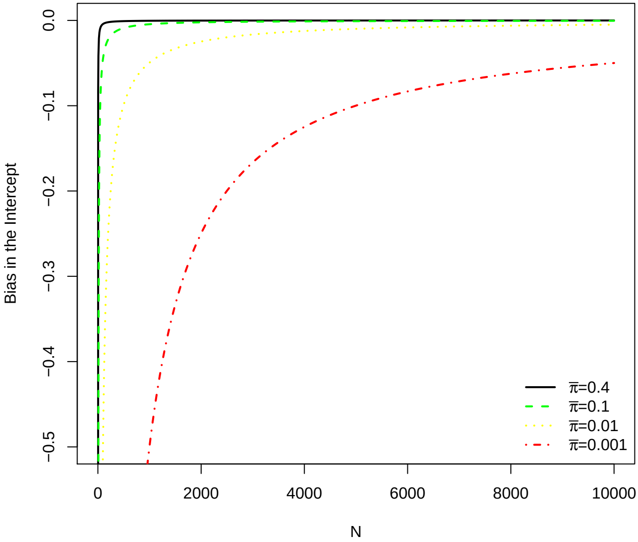
where $\bar{\pi} = \overline{\Pr(Y = 1)}$ is < 0.5.

Bias is:

- always negative,
- worse as $\bar{\pi} \rightarrow 0$ (for fixed N),
- disappearing as $N \rightarrow \infty$.

Implication: *Logit/probit "work best" around* $\bar{\pi} = 0.5$.

Rare Event Bias, Illustrated



Alternatively, we can:

- Calculate $\tau = \frac{N_1 s}{N}$
- Collect data on all "1s"
- Sample from the "0s"
- Estimate a logit*
- *Correct* the estimates ex post...

* Yes, it *has* to be a logit, not a probit or something else...

Sampling...

- $\tau =$ fraction of "1s" in the population
- $\bar{Y} =$ fraction of '1s" in the sample
- K&Z suggest $\bar{Y} \in [0.2, 0.5]$

Weighting...

$$w_1 = \frac{\tau}{\bar{Y}} \text{ (weights for "1s")}$$

$$w_0 = \frac{1 - \tau}{1 - \bar{Y}} \text{ (weights for "0s")}$$

$$\ln L(\beta|Y) = \sum_{i=1}^{N} w_1 Y_i \ln \Lambda(\mathbf{X}_i \boldsymbol{\beta}) + w_0(1 - Y_i) \ln[1 - \Lambda(\mathbf{X}_i \boldsymbol{\beta})]$$

Weighting:

- Good under (possible) misspecification, but
- Not as efficient as "prior correction," and
- Gets s.e.s wrong...

Case-Control Data: Prior Correction

The prior correction is:

$$\hat{\beta}_{0\text{pc}} = \hat{\beta}_0 - \ln\left[\left(\frac{1-\tau}{\tau}\right)\left(\frac{\bar{Y}}{1-\bar{Y}}\right)\right]$$

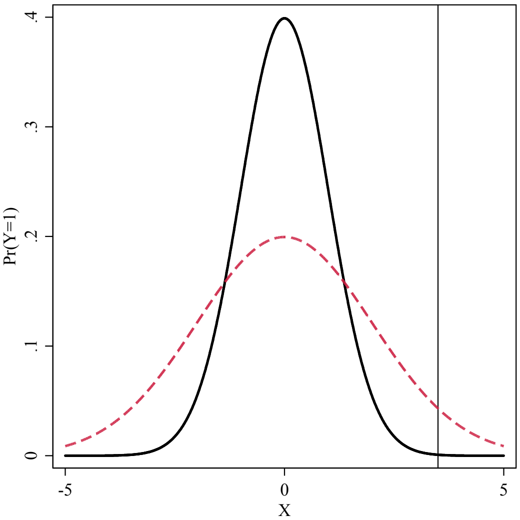
$$\text{bias}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\xi$$

where $\xi = f[w_i, \hat{\pi}_i, \mathbf{X}]$.

Correction is

$$\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}} - \text{bias}(\hat{\boldsymbol{\beta}})$$

- Bias correction introduces additional variability...
- Ignoring it yields underpredictions (again).


.4
.3
.2
.1
0
Pr(Y=1)
-5
0
5
X

Post-Correction Adjustments

Use:

$$\Pr(Y_i = 1) \approx \tilde{\pi}_i + C_i$$

where

$$C_i = (0.5 - \tilde{\pi}_i)\tilde{\pi}_i(1 - \tilde{\pi}_i)\mathbf{X}_i\mathbf{V}(\tilde{\beta})\mathbf{X}_i'$$

Puhr et al. (2017) note that Firth's method indices bias (toward 0.5) in predicted probabilities, and that the bias is worse when the baseline $\Pr(Y_i = 1)$ is low.

They introduce two modifications to deal with this:

- "Firth's logit with intercept correction" (FLIC)
- "Firth's logit with added covariate" (FLAC)

Through simulations, they show that both remove the bias; they have a slight preference for FLAC, but note that both work well relative to unmodified Firth regression. See also here for a more recent analysis of using Firth's methods for rare events.

Different data:

- Washington University's American Panel Study (TAPS)
- $N \approx 1000$ U.S. respondents, 2012-2017
- Outcome: "During the past year, have you ever run out of gas while driving a car or other vehicle?" (`RunOutOfGas`; 0=no, 1=yes)
- Predictors:
  - `Education` – twelve-category ordinal variable with values ranging from 3 to 15;
  - `Income` – a 15-category ordinal variable (each unit roughly corresponds to an increase of $10,000 in annual income);
  - `Age` in years, as of 2016 (divided by 10);
  - `Female` – a binary indicator of sex, naturally-coded;
  - Racial classifications – binary variables for `White`, `Black`, and `Asian` identification;
  - Binary political party variables for `Democrat` and `GOP`; and
  - `Ideology` – a seven-point Likert variable, higher values indicate greater political conservatism

Basic Logit...

```
> table(TAPS$RunOutOfGas)

  0   1 
943  28 

> prop.table(table(TAPS$RunOutOfGas))

     0      1 
0.9712 0.0288 

> ROGlogit<-glm(RunOutOfGas~Education+Age10+Female+White+Black+Asian+
+                       Democrat+GOP+Ideology,data=TAPS,family=binomial)

> summary(ROGlogit)

Deviance Residuals: 
   Min      1Q  Median      3Q     Max  
-0.661  -0.248  -0.206  -0.170   2.962  

Coefficients:
            Estimate Std. Error z value Pr(>|z|)  
(Intercept)  -1.9347     1.8114   -1.07    0.285  
Education    -0.1185     0.1118   -1.06    0.289  
Age10        -0.2107     0.1341   -1.57    0.116  
Female        0.2911     0.3966    0.73    0.463  
White         0.4348     0.7260    0.60    0.549  
Black         1.3503     0.7602    1.78    0.076 .
Asian         1.8616     0.8717    2.14    0.033 *
Democrat      0.2743     0.4999    0.55    0.583  
GOP          -0.3170     0.5926   -0.53    0.593  
Ideology      0.0217     0.1097    0.20    0.843  
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1   1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 253.77  on 970  degrees of freedom
Residual deviance: 238.13  on 961  degrees of freedom
AIC: 258.1
```

King/Zeng Rare Events Logit...

```
> devtools::install_github('IQSS/Zelig')
> library(Zelig)

> RElogit<-relogit(RunOutOfGas~Education+Age10+Female+White+Black+Asian+
+                  Democrat+GOP+Ideology, tau = NULL,
+                  case.control = c("prior", "weighting"), # n/a here
+                  bias.correct = TRUE, data=TAPS)

> summary(RElogit)

Call:
relogit(formula = RunOutOfGas ~ Education + Age10 + Female +
    White + Black + Asian + Democrat + GOP + Ideology, data = TAPS,
    tau = NULL, bias.correct = TRUE, case.control = c("prior",
        "weighting"))

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.7875     1.8114   -0.99    0.324
Education    -0.1173     0.1118   -1.05    0.294
Age10        -0.2076     0.1341   -1.55    0.122
Female        0.2737     0.3966    0.69    0.490
White         0.3916     0.7260    0.54    0.590
Black         1.3600     0.7602    1.79    0.074 .
Asian         1.9579     0.8717    2.25    0.025 *
Democrat      0.2549     0.4999    0.51    0.610
GOP          -0.3057     0.5926   -0.52    0.606
Ideology      0.0272     0.1097    0.25    0.804
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 253.77  on 970  degrees of freedom
Residual deviance: 238.13  on 961  degrees of freedom
AIC: 258.1

Number of Fisher Scoring iterations: 7
```

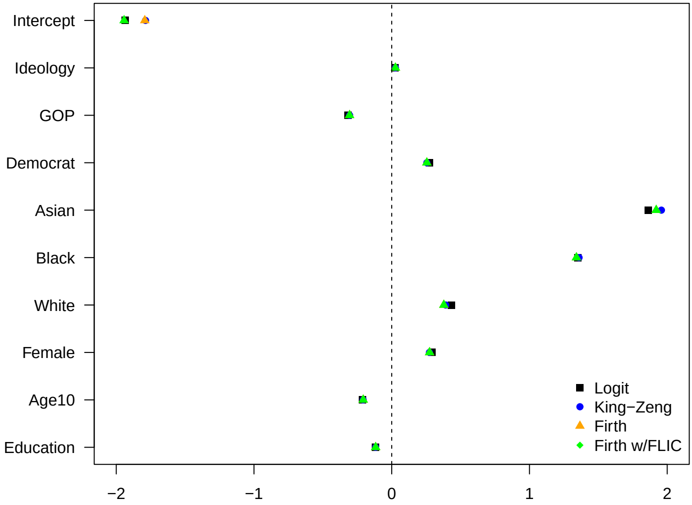
Firth Logit (for comparison)

```
> relogit.firth<-logistf(RunOutOfGas~Education+Age10+Female+White+Black+Asian+
+                       Democrat+GOP+Ideology,data=TAPS)

> summary(relogit.firth)

logistf(formula = RunOutOfGas ~ Education + Age10 + Female +
    White + Black + Asian + Democrat + GOP + Ideology, data = TAPS)

Model fitted by Penalized ML
Coefficients:
               coef se(coef) lower 0.95 upper 0.95  Chisq      p method
(Intercept) -1.7929    1.657     -5.362     1.6045 1.0457 0.3065      2
Education   -0.1167    0.103     -0.331     0.1009 1.1154 0.2909      2
Age10       -0.2071    0.124     -0.469     0.0498 2.4952 0.1142      2
Female       0.2749    0.367     -0.478     1.0490 0.5124 0.4741      2
White        0.3782    0.646     -1.007     1.7513 0.2769 0.5987      2
Black        1.3409    0.677     -0.182     2.7141 2.9875 0.0839      2
Asian        1.9202    0.766      0.149     3.4429 4.4610 0.0347      2
Democrat     0.2550    0.464     -0.688     1.2418 0.2767 0.5989      2
GOP         -0.3061    0.546     -1.479     0.7889 0.2969 0.5858      2
Ideology     0.0267    0.101     -0.191     0.2333 0.0613 0.8044      2

Method: 1-Wald, 2-Profile penalized log-likelihood, 3-None

Likelihood ratio test=17.5 on 9 df, p=0.0415, n=971
Wald test = 318 on 9 df, p = 0
```

Firth Logit with FLIC

```
> relogit.flic<-logistf(RunOutOfGas~Education+Age10+Female+White+Black+Asian+
+                           Democrat+GOP+Ideology,data=TAPS,flic=TRUE)

> summary(relogit.flic)

logistf(formula = RunOutOfGas ~ Education + Age10 + Female +
    White + Black + Asian + Democrat + GOP + Ideology, data = TAPS,
    flic = TRUE)

Model fitted by Penalized ML
Coefficients:
               coef se(coef) lower 0.95 upper 0.95  Chisq      p method
(Intercept) -1.9430    1.807     -5.486     1.5995 1.0457 0.3065      1
Education   -0.1167    0.112     -0.331     0.1009 1.1154 0.2909      2
Age10       -0.2071    0.134     -0.469     0.0498 2.4952 0.1142      2
Female       0.2749    0.397     -0.478     1.0490 0.5124 0.4741      2
White        0.3782    0.720     -1.007     1.7513 0.2769 0.5987      2
Black        1.3409    0.756     -0.182     2.7141 2.9875 0.0839      2
Asian        1.9202    0.857      0.149     3.4429 4.4610 0.0347      2
Democrat     0.2550    0.501     -0.688     1.2418 0.2767 0.5989      2
GOP         -0.3061    0.590     -1.479     0.7889 0.2969 0.5858      2
Ideology     0.0267    0.110     -0.191     0.2333 0.0613 0.8044      2

Method: 1-Wald, 2-Profile penalized log-likelihood, 3-None

Likelihood ratio test=17.5 on 9 df, p=0.0415, n=971
Wald test = 299 on 9 df, p = 0
```

Summarizing: $\hat{\beta}$s



Some Final Thoughts

In no particular order:

- The key to doing King-Zeng is to be able to conduct C-C sampling *in advance*
- BUT: The R implementation of K&Z (in `Zelig`) is currently a bit buggy (its dependencies are all messed up...)
- In practice: the Firth + FLIC approach is generally superior to King/Zeng (and arguably should *always* be used for binary-response regressions, especially with small-to-medium Ns)
- Also: Remember that as your N gets big, the problem goes away; Paul Allision has a (old, but useful) blog post on that topic.

Other Binary-Response Extensions

Things we'll talk about later:

- Binary responses in panel / longitudinal data
- Multilevel / hierarchical models for binary responses
- Models with (binary) sample selection
- Measurement models for binary outcomes (e.g., item response models)

Things we won't talk about:

- Semi- and non-parametric models (e.g., Klen and Spady (1993), Christensen (2024))
- "Heteroscedastic" models (where $\sigma_i^2 \neq \sigma^2 \, \forall \, i$) (see, e.g., Alvarez and Brehm 1995, 1997; Tutz 2018)
- "Bivariate" models, where (e.g.):

$$\{Y_{1i}, Y_{2i}\} \sim BVN(0, 0, 1, 1, \rho)$$

(e.g. Marra et al. 2023)

Nominal Outcomes

Motivation: Discrete *Outcomes*

For J possible outcomes $j \in \{1, 2, ...J\}$ on Y:

$$\Pr(Y_i = j) = P_{ij}$$

we require that:

$$\sum_{j=1}^{J} P_{ij} = 1.$$

To ensure $P_{ij} > 0$:

$$P_{ij} = \exp(\mathbf{X}_i \boldsymbol{\beta}_j).$$

Rescale to make $0 < P_{ij} < 1$:

$$\Pr(Y_i = j) \equiv P_{ij} = \frac{\exp(\mathbf{X}_i \boldsymbol{\beta}_j)}{\sum_{j=1}^{J} \exp(\mathbf{X}_i \boldsymbol{\beta}_j)}$$

This ensures that:

- $\Pr(Y_i = j) \in (0, 1)$
- $\sum_{j=1}^{J} \Pr(Y_i = j) = 1.0$

Constrain $\boldsymbol{\beta}_1 = \mathbf{0}$; then:

$$\Pr(Y_i = 1) = \frac{1}{1 + \sum_{j=2}^{J} \exp(\mathbf{X}_i \boldsymbol{\beta}_j')}$$

$$\Pr(Y_i = j) = \frac{\exp(\mathbf{X}_i \boldsymbol{\beta}_j')}{1 + \sum_{j=2}^{J} \exp(\mathbf{X}_i \boldsymbol{\beta}_j')}$$

where $\boldsymbol{\beta}_j' = \boldsymbol{\beta}_j - \boldsymbol{\beta}_1$.

This is *multinomial logit*.

Alternative Motivation: Discrete *Choice*

Utility function for i over j alternatives:

$$U_{ij} = \mu_i + \epsilon_{ij}$$

... varues with $\mathbf{X}$:

$$\mu_i = \mathbf{X}_i \boldsymbol{\beta}_j$$

Yields:

$$\begin{aligned}
\Pr(Y_i = j) &= \Pr(U_{ij} > U_{i\ell} \,\forall\, \ell \neq j \in J) \\
&= \Pr(\mu_i + \epsilon_{ij} > \mu_i + \epsilon_{i\ell} \,\forall\, \ell \neq j \in J) \\
&= \Pr(\mathbf{X}_i \boldsymbol{\beta}_j + \epsilon_{ij} > \mathbf{X}_i \boldsymbol{\beta}_\ell + \epsilon_{i\ell} \,\forall\, \ell \neq j \in J) \\
&= \Pr(\epsilon_{ij} - \epsilon_{i\ell} > \mathbf{X}_i \boldsymbol{\beta}_\ell - \mathbf{X}_i \boldsymbol{\beta}_j \,\forall\, \ell \neq j \in J)
\end{aligned}$$

$\epsilon \sim$???

- *Type I Extreme Value*
- Density: $f(\epsilon) = \exp[-\epsilon - \exp(-\epsilon)]$
- CDF: $\int f(\epsilon) \equiv F(\epsilon) = \exp[-\exp(-\epsilon)]$
- $\rightarrow$ Multinomial Logit

The probability of choosing choice j is:

$$
\begin{aligned}
\Pr(Y_i = j) &= \Pr(U_j > U_1, U_j > U_2, ...U_j > U_J) \\
&= \int f(\epsilon_j) \left[\int_{-\infty}^{\epsilon_{ij} + \mathbf{X}_i \boldsymbol{\beta}_j - \mathbf{X}_i \boldsymbol{\beta}_1} f(\epsilon_1) d\epsilon_1 \times \int_{-\infty}^{\epsilon_{ij} + \mathbf{X}_i \boldsymbol{\beta}_j - \mathbf{X}_i \boldsymbol{\beta}_2} f(\epsilon_2) d\epsilon_2 \times ... \right] d\epsilon_j \\
&= \int f(\epsilon_j) \times \exp[-\exp(\epsilon_{ij} + \mathbf{X}_i \boldsymbol{\beta}_j - \mathbf{X}_i \boldsymbol{\beta}_1)] \times \\
& \quad \exp[-\exp(\epsilon_{ij} + \mathbf{X}_i \boldsymbol{\beta}_j - \mathbf{X}_i \boldsymbol{\beta}_2)] \times ... d\epsilon_j \\
&= \frac{\exp(\mathbf{X}_i \boldsymbol{\beta}_j)}{\sum_{j=1}^{J} \exp(\mathbf{X}_i \boldsymbol{\beta}_j)}
\end{aligned}
$$

Define:

$$\begin{aligned}\delta_{ij} &= 1 \text{ if } Y_i = j, \\ &= 0 \text{ otherwise.}\end{aligned}$$

Then:

$$\begin{aligned}L_i &= \prod_{j=1}^{J} [\Pr(Y_i = j)]^{\delta_{ij}} \\ &= \prod_{j=1}^{J} \left[\frac{\exp(\mathbf{X}_i \boldsymbol{\beta}_j)}{\sum_{j=1}^{J} \exp(\mathbf{X}_i \boldsymbol{\beta}_j)} \right]^{\delta_{ij}}\end{aligned}$$

So:

$$L = \prod_{i=1}^{N} \prod_{j=1}^{J} \left[\frac{\exp(\mathbf{X}_i \boldsymbol{\beta}_j)}{\sum_{j=1}^{J} \exp(\mathbf{X}_i \boldsymbol{\beta}_j)} \right]^{\delta_{ij}}$$

and:

$$\ln L = \sum_{i=1}^{N} \sum_{j=1}^{J} \delta_{ij} \ln \left[\frac{\exp(\mathbf{X}_i \boldsymbol{\beta}_j)}{\sum_{j=1}^{J} \exp(\mathbf{X}_i \boldsymbol{\beta}_j)} \right]$$

...which can be estimated in the usual way(s).

Example: The 1992 U.S. Presidential Election

1992 American National Election Study

Data:

- Y (PresVote) $\in \{\text{Bush}(=1), \text{Clinton}(=2), \text{Perot}(=3)\}$
- $\mathbf{X}$ = political demographic characteristics + "feeling thermometers"

```
> describe(NES92)
           vars    n    mean      sd median trimmed     mad  min  max range  skew kurtosis    se
ID            1 1473 4671.15 1104.02   5113 4681.28 1546.35 3001 6251  3250 -0.11    -1.66 28.77
VotedFor*     2 1473    1.85    0.71      2    1.82    1.48    1    3     2  0.22    -1.03  0.02
PresVote      3 1473    1.85    0.71      2    1.82    1.48    1    3     2  0.22    -1.03  0.02
PartyID       4 1473    3.75    2.11      3    3.69    2.97    1    7     6  0.15    -1.39  0.06
Age           5 1473   45.89   16.67     43   44.85   17.79   18   91    73  0.50    -0.72  0.43
FamIncome     6 1473   15.53    5.76     16   16.10    5.93    1   24    23 -0.78    -0.17  0.15
Female        7 1473    0.51    0.50      1    0.52    0.00    0    1     1 -0.06    -2.00  0.01
White*        8 1473    1.88    0.33      2    1.97    0.00    1    2     1 -2.31     3.36  0.01
FT.Bush       9 1473   51.75   27.26     60   52.99   29.65    0  100   100 -0.30    -0.72  0.71
FT.Clinton   10 1473   55.77   25.08     60   57.35   29.65    0  100   100 -0.45    -0.37  0.65
FT.Perot     11 1473   44.85   26.51     50   44.90   29.65    0  100   100 -0.16    -0.68  0.69
```

Model:

$$\texttt{PresVote}_i = f(\beta_0+\beta_1\times\texttt{PartyID}_i+\beta_2\times\texttt{Age}_i+\beta_3\times\texttt{White}_i+\beta_4\times\texttt{Female}_i)$$

MNL #1, using `vglm` ("Baseline" = Perot)

```
> NES92.mlogit<-vglm(PresVote~PartyID+Age+White+Female,multinomial,data=NES92)
> summary(NES92.mlogit)

Call:
vglm(formula = PresVote ~ PartyID + Age + White + Female, family = multinomial,
    data = NES92)

Coefficients:
              Estimate Std. Error z value  Pr(>|z|)
(Intercept):1 -1.98008    0.52454   -3.77   0.00016 ***
(Intercept):2  3.82657    0.46402    8.25   < 2e-16 ***
PartyID:1      0.50132    0.04870   10.29   < 2e-16 ***
PartyID:2     -0.63429    0.04918  -12.90   < 2e-16 ***
Age:1          0.01556    0.00504    3.09   0.00203 **
Age:2          0.01296    0.00510    2.54   0.01096 *
WhiteWhite:1  -0.87918    0.43605   -2.02   0.04377 *
WhiteWhite:2  -1.86826    0.38611   -4.84 0.0000013 ***
Female:1       0.50928    0.16266    3.13   0.00174 **
Female:2       0.38427    0.16267    2.36   0.01816 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,1]/mu[,3]), log(mu[,2]/mu[,3])
Residual deviance: 2107 on 2936 degrees of freedom
Log-likelihood: -1054 on 2936 degrees of freedom
Number of Fisher scoring iterations: 5
No Hauck-Donner effect found in any of the estimates
Reference group is level  3  of the response
```

MNL #2, using `multinom` ("Baseline" = Perot)

```
> NES92$PresVote2<-factor(NES92$PresVote,
+                         levels = c("3", "1", "2"),
+                         labels = c("Perot", "Bush", "Clinton"))

> NES92.mlogit2<-multinom(PresVote2~PartyID+Age+White+Female,data=NES92)
# weights:  18 (10 variable)
initial  value 1618.255901
iter  10 value 1080.908630
final  value 1053.650588
converged

> summary(NES92.mlogit2)
Call:
multinom(formula = PresVote2 ~ PartyID + Age + White + Female,
    data = NES92)

Coefficients:
        (Intercept) PartyID    Age WhiteWhite Female
Bush          -1.98   0.501 0.0156     -0.879  0.509
Clinton        3.83  -0.634 0.0130     -1.868  0.384

Std. Errors:
        (Intercept) PartyID     Age WhiteWhite Female
Bush          0.525  0.0487 0.00504      0.436  0.163
Clinton       0.464  0.0492 0.00510      0.386  0.163

Residual Deviance: 2107
AIC: 2127
```

First, we have to "reshape" the data:

```
> head(NES92)
    ID VotedFor PresVote PartyID Age FamIncome Female     White FT.Bush FT.Clinton FT.Perot PresVote2
1 3001     Bush        1       6  31        20      0     White      85         30        0      Bush
2 3002     Bush        1       7  89         9      1     White     100          0        0      Bush
3 3003     Bush        1       7  35        17      1     White      85         30       60      Bush
4 3005  Clinton        2       6  27         3      1 Non-White      40         60       60   Clinton
5 3006  Clinton        2       2  54        15      1     White      30         70       50   Clinton
6 3007  Clinton        2       1  45         2      1 Non-White      15         70       50   Clinton

> AltNES92<-dfidx(NES92,varying=9:11,shape="wide",choice="VotedFor")

> head(AltNES92)
~~~~~~~
 first 10 observations out of 4419
~~~~~~~
     ID VotedFor PresVote PartyID Age FamIncome Female     White PresVote2  FT    idx
1  3001     TRUE        1       6  31        20      0     White      Bush  85 1:Bush
2  3001    FALSE        1       6  31        20      0     White      Bush  30 1:nton
3  3001    FALSE        1       6  31        20      0     White      Bush   0 1:erot
4  3002     TRUE        1       7  89         9      1     White      Bush 100 2:Bush
5  3002    FALSE        1       7  89         9      1     White      Bush   0 2:nton
6  3002    FALSE        1       7  89         9      1     White      Bush   0 2:erot
7  3003     TRUE        1       7  35        17      1     White      Bush  85 3:Bush
8  3003    FALSE        1       7  35        17      1     White      Bush  30 3:nton
9  3003    FALSE        1       7  35        17      1     White      Bush  60 3:erot
```

Now, fit the model:

```
> NES92.mlogit3<-mlogit(VotedFor~0|PartyID+Age+White+Female,data=AltNES92,reflevel="Perot")
> summary(NES92.mlogit3)

Call:
mlogit(formula = VotedFor ~ 0 | PartyID + Age + White + Female,
    data = AltNES92, reflevel = "Perot", method = "nr")

Frequencies of alternatives:choice
  Perot    Bush Clinton
  0.191   0.339   0.469

nr method
5 iterations, 0h:0m:0s
g'(-H)^-1g = 4.94E-08
gradient close to zero

Coefficients :
                    Estimate Std. Error z-value Pr(>|z|)
(Intercept):Bush    -1.98008    0.52454   -3.77  0.00016 ***
(Intercept):Clinton  3.82657    0.46403    8.25  2.2e-16 ***
PartyID:Bush         0.50132    0.04870   10.29  < 2e-16 ***
PartyID:Clinton     -0.63429    0.04918  -12.90  < 2e-16 ***
Age:Bush             0.01556    0.00504    3.09  0.00203 **
Age:Clinton          0.01296    0.00510    2.54  0.01096 *
WhiteWhite:Bush     -0.87918    0.43606   -2.02  0.04378 *
WhiteWhite:Clinton  -1.86826    0.38612   -4.84  1.3e-06 ***
Female:Bush          0.50928    0.16266    3.13  0.00174 **
Female:Clinton       0.38427    0.16267    2.36  0.01816 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Log-Likelihood: -1050
McFadden R^2:  0.311
Likelihood ratio test : chisq = 952 (p.value = <2e-16)
```

MNL: 1992 Election ("Baseline" = Bush)

```
> Bush.nes92.mlogit<-vglm(PresVote~PartyID+Age+White+Female,
+                         data=NES92,family=multinomial(refLevel=1))
> summary(Bush.nes92.mlogit)

Call:
vglm(formula = PresVote ~ PartyID + Age + White + Female, family = multinomial(refLevel = 1),
    data = NES92)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept):1  5.80665    0.44301   13.11  < 2e-16 ***
(Intercept):2  1.98008    0.52454    3.77  0.00016 ***
PartyID:1     -1.13561    0.05486  -20.70  < 2e-16 ***
PartyID:2     -0.50132    0.04870  -10.29  < 2e-16 ***
Age:1         -0.00260    0.00514   -0.51  0.61276
Age:2         -0.01556    0.00504   -3.09  0.00203 **
WhiteWhite:1  -0.98908    0.31346   -3.16  0.00160 **
WhiteWhite:2   0.87918    0.43605    2.02  0.04377 *
Female:1      -0.12500    0.16895   -0.74  0.45936
Female:2      -0.50928    0.16266   -3.13  0.00174 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,2]/mu[,1]), log(mu[,3]/mu[,1])
Residual deviance: 2107 on 2936 degrees of freedom
Log-likelihood: -1054 on 2936 degrees of freedom
Number of Fisher scoring iterations: 5
No Hauck-Donner effect found in any of the estimates
Reference group is level  1  of the response
```

MNL: 1992 Election ("Baseline" = Clinton)

```
> Clinton.nes92.mlogit<-vglm(PresVote~PartyID+Age+White+Female,
+                         data=NES92,family=multinomial(refLevel=2))
> summary(Clinton.nes92.mlogit)

Call:
vglm(formula = PresVote ~ PartyID + Age + White + Female, family = multinomial(refLevel = 2),
    data = NES92)

Coefficients:
              Estimate Std. Error z value  Pr(>|z|)
(Intercept):1 -5.80665    0.44301  -13.11   < 2e-16 ***
(Intercept):2 -3.82657    0.46402   -8.25   < 2e-16 ***
PartyID:1      1.13561    0.05486   20.70   < 2e-16 ***
PartyID:2      0.63429    0.04918   12.90   < 2e-16 ***
Age:1          0.00260    0.00514    0.51    0.6128
Age:2         -0.01296    0.00510   -2.54    0.0110 *
WhiteWhite:1   0.98908    0.31346    3.16    0.0016 **
WhiteWhite:2   1.86826    0.38611    4.84 0.0000013 ***
Female:1       0.12500    0.16895    0.74    0.4594
Female:2      -0.38427    0.16267   -2.36    0.0182 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Names of linear predictors: log(mu[,1]/mu[,2]), log(mu[,3]/mu[,2])
Residual deviance: 2107 on 2936 degrees of freedom
Log-likelihood: -1054 on 2936 degrees of freedom
Number of Fisher scoring iterations: 5
Warning: Hauck-Donner effect detected in the following estimate(s):
'(Intercept):2'
Reference group is level  2  of the response
```

PartyID Coefficient Estimates and "Baselines"

Note: PartyID is 1 (strong Democrat) $\rightarrow$ 7 (strong Republican)

		"Baseline" category		
		Clinton	Perot	Bush
Comparison	Clinton	–	-0.63	-1.14
Category	Perot	0.63	–	-0.50
	Bush	1.14	0.50	–

Consider the choice of Bush vs. Perot:

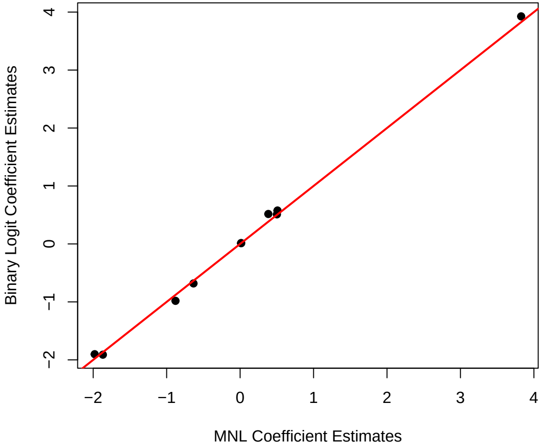
```
> NES92$PickBush<-NA
> NES92$PickBush<-ifelse(NES92$VotedFor=="Bush",1,NES92$PickBush)
> NES92$PickBush<-ifelse(NES92$VotedFor=="Perot",0,NES92$PickBush)
> BushBinary<-glm(PickBush~PartyID+Age+White+Female,data=NES92,family="binomial")
> summary(BushBinary)

Call:
glm(formula = PickBush ~ PartyID + Age + White + Female, family = "binomial",
    data = NES92)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.9024     0.5372   -3.54  0.00040 ***
PartyID       0.5106     0.0505   10.12  < 2e-16 ***
Age           0.0143     0.0052    2.75  0.00595 **
WhiteWhite   -0.9817     0.4586   -2.14  0.03230 *
Female        0.5768     0.1683    3.43  0.00061 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1022.50  on 781  degrees of freedom
Residual deviance:  880.28  on 777  degrees of freedom
  (691 observations deleted due to missingness)
AIC: 890.3

Number of Fisher Scoring iterations: 4
```

What about Clinton vs. Perot?:

```
> NES92$PickClinton<-NA
> NES92$PickClinton<-ifelse(NES92$VotedFor=="Clinton",1,NES92$PickClinton)
> NES92$PickClinton<-ifelse(NES92$VotedFor=="Perot",0,NES92$PickClinton)
> ClintonBinary<-glm(PickClinton~PartyID+Age+White+Female,data=NES92,family="binomial")
> summary(ClintonBinary)

Call:
glm(formula = PickClinton ~ PartyID + Age + White + Female, family = "binomial",
    data = NES92)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.92614    0.48490    8.10  5.6e-16 ***
PartyID     -0.68125    0.05301  -12.85  < 2e-16 ***
Age          0.01381    0.00537    2.57   0.0101 *
WhiteWhite  -1.91056    0.39879   -4.79  1.7e-06 ***
Female       0.51690    0.17024    3.04   0.0024 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1171.48  on 972  degrees of freedom
Residual deviance:  861.57  on 968  degrees of freedom
  (500 observations deleted due to missingness)
AIC: 871.6

Number of Fisher Scoring iterations: 5
```

MNL and Binary Logit (continued)

Are the $\hat{\beta}$s the same? (A: Yes, basically...)



It is exactly the same as the multinomial logit model. Period.

Choice-Specific Covariates: Data Structure

```
> head(AltNES92)

      ID VotedFor PresVote PartyID   Age FamIncome Female White     PresVote2    FT idx$id1 $id2
 * <dbl> <lgl>       <dbl>   <dbl> <dbl>     <dbl>  <dbl> <chr>     <fct>     <dbl>   <int> <fct>
 1  3001 TRUE            1       6    31        20      0 White     Bush         85       1 Bush
 2  3001 FALSE           1       6    31        20      0 White     Bush         30       1 Clinton
 3  3001 FALSE           1       6    31        20      0 White     Bush          0       1 Perot
 4  3002 TRUE            1       7    89         9      1 White     Bush        100       2 Bush
 5  3002 FALSE           1       7    89         9      1 White     Bush          0       2 Clinton
 6  3002 FALSE           1       7    89         9      1 White     Bush          0       2 Perot
 7  3003 TRUE            1       7    35        17      1 White     Bush         85       3 Bush
 8  3003 FALSE           1       7    35        17      1 White     Bush         30       3 Clinton
 9  3003 FALSE           1       7    35        17      1 White     Bush         60       3 Perot
10  3005 FALSE           2       6    27         3      1 Non-White Clinton      40       4 Bush
# 4,409 more rows
# Use 'print(n = ...)' to see more rows
```

Note that:

$$\Pr(Y_{ij} = 1) = \frac{\exp(\mathbf{Z}_{ij}\gamma)}{\sum_{j=1}^{J} \exp(\mathbf{Z}_{ij}\gamma)}$$

Combinations: $\mathbf{X}_i\boldsymbol{\beta}$ and $\mathbf{Z}_{ij}\gamma$:

- "Fixed effects" (choice-specific intercepts), plus
- Observation-specific **X**s, plus
- Interactions...

CL in R (Feeling Thermometers only)

```
> NES92.clogit<-mlogit(VotedFor~FT,data=AltNES92,reflevel="Perot")
> summary(NES92.clogit)

Call:
mlogit(formula = VotedFor ~ FT, data = AltNES92, reflevel = "Perot",
    method = "nr")

Frequencies of alternatives:choice
  Perot    Bush Clinton
  0.191   0.339   0.469

nr method
6 iterations, 0h:0m:0s
g'(-H)^-1g = 0.00219
successive function values within tolerance limits

Coefficients :
                    Estimate Std. Error z-value   Pr(>|z|)
(Intercept):Bush     0.03307    0.10039    0.33       0.74
(Intercept):Clinton  0.45841    0.09253    4.95 0.00000073 ***
FT                   0.07512    0.00314   23.89    < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -801
McFadden R^2:  0.476
Likelihood ratio test : chisq = 1460 (p.value = <2e-16)
```

CL in R ("Full" Model)

```
> NES92.clogit2<-mlogit(VotedFor~FT|PartyID+Age+White+Female,data=AltNES92,reflevel="Perot")
> summary(NES92.clogit2)

Frequencies of alternatives:choice
  Perot    Bush Clinton
  0.191   0.339   0.469

nr method
6 iterations, 0h:0m:0s
g'(-H)^-1g = 3.06E-08
gradient close to zero

Coefficients :
                     Estimate Std. Error z-value Pr(>|z|)
(Intercept):Bush     -0.39416    0.58730   -0.67  0.50214
(Intercept):Clinton   2.69235    0.50403    5.34  9.2e-08 ***
FT                    0.06231    0.00325   19.17  < 2e-16 ***
PartyID:Bush          0.22298    0.05842    3.82  0.00014 ***
PartyID:Clinton      -0.40620    0.05798   -7.01  2.4e-12 ***
Age:Bush              0.00868    0.00612    1.42  0.15639
Age:Clinton           0.00839    0.00598    1.40  0.16040
WhiteWhite:Bush      -1.31961    0.47725   -2.77  0.00569 **
WhiteWhite:Clinton   -1.57156    0.41404   -3.80  0.00015 ***
Female:Bush           0.39271    0.19995    1.96  0.04953 *
Female:Clinton        0.28585    0.19474    1.47  0.14213
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -723
McFadden R^2:  0.527
Likelihood ratio test : chisq = 1610 (p.value = <2e-16)
```

Interpretation: Baseline MNL Results

```
> NES.MNL<-vglm(PresVote~PartyID+Age+White+Female,data=NES92,
+               multinomial(refLevel=1)) # Bush is comparison category
> summaryvglm(NES.MNL)

Call:
vglm(formula = PresVote ~ PartyID + Age + White + Female, family = multinomial(refLevel = 1),
    data = NES92)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept):1  5.80665    0.44301   13.11  < 2e-16 ***
(Intercept):2  1.98008    0.52454    3.77  0.00016 ***
PartyID:1     -1.13561    0.05486  -20.70  < 2e-16 ***
PartyID:2     -0.50132    0.04870  -10.29  < 2e-16 ***
Age:1         -0.00260    0.00514   -0.51  0.61276
Age:2         -0.01556    0.00504   -3.09  0.00203 **
WhiteWhite:1  -0.98908    0.31346   -3.16  0.00160 **
WhiteWhite:2   0.87918    0.43605    2.02  0.04377 *
Female:1      -0.12500    0.16895   -0.74  0.45936
Female:2      -0.50928    0.16266   -3.13  0.00174 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Names of linear predictors: log(mu[,2]/mu[,1]), log(mu[,3]/mu[,1])
Residual deviance: 2107 on 2936 degrees of freedom
Log-likelihood: -1054 on 2936 degrees of freedom
Number of Fisher scoring iterations: 5
No Hauck-Donner effect found in any of the estimates
Reference group is level  1  of the response
```

Pretty Things Using `modelsummary`

```
> mod<-list(NES92.mlogit2)

> get_estimates(mod[[1]])
          term estimate std.error conf.level conf.low conf.high statistic df.error  p.value response s.value group
1  (Intercept)  -1.9797   0.52455       0.95 -3.00781   -0.9516     -3.77      Inf 1.61e-04     Bush    12.6
2      PartyID   0.5013   0.04870       0.95  0.40586    0.5967     10.29      Inf 7.47e-25     Bush    80.1
3          Age   0.0156   0.00504       0.95  0.00568    0.0254      3.09      Inf 2.03e-03     Bush     8.9
4        White  -0.8794   0.43607       0.95 -1.73404   -0.0247     -2.02      Inf 4.37e-02     Bush     4.5
5       Female   0.5093   0.16266       0.95  0.19047    0.8281      3.13      Inf 1.74e-03     Bush     9.2
6  (Intercept)   3.8267   0.46404       0.95  2.91718    4.7362      8.25      Inf 1.63e-16  Clinton    52.4
7      PartyID  -0.6343   0.04918       0.95 -0.73067   -0.5379    -12.90      Inf 4.71e-38  Clinton   124.0
8          Age   0.0130   0.00510       0.95  0.00297    0.0229      2.54      Inf 1.10e-02  Clinton     6.5
9        White  -1.8683   0.38613       0.95 -2.62512   -1.1115     -4.84      Inf 1.31e-06  Clinton    19.5
10      Female   0.3843   0.16267       0.95  0.06544    0.7031      2.36      Inf 1.82e-02  Clinton     5.8
```

```
> modelsummary(mod,shape=term+response~statistic)
```

		(1)	
	response	Est.	S.E.
(Intercept)	Bush	-1.980	0.525
	Clinton	3.827	0.464
PartyID	Bush	0.501	0.049
	Clinton	-0.634	0.049
Age	Bush	0.016	0.005
	Clinton	0.013	0.005
WhiteWhite	Bush	-0.879	0.436
	Clinton	-1.868	0.386
Female	Bush	0.509	0.163
	Clinton	0.384	0.163
Num.Obs.		1473	
R2		0.311	
R2 Adj.		0.310	
AIC		2127.3	
BIC		2180.3	
RMSE		0.37	

```
> modelsummary(mod,shape=model+term~response)
```

		Bush	Clinton
(1)	(Intercept)	-1.980	3.827
		(0.525)	(0.464)
	PartyID	0.501	-0.634
		(0.049)	(0.049)
	Age	0.016	0.013
		(0.005)	(0.005)
	WhiteWhite	-0.879	-1.868
		(0.436)	(0.386)
	Female	0.509	0.384
		(0.163)	(0.163)

Global $\ln LR$ statistic Q tests:

$$\hat{\boldsymbol{\beta}} = \mathbf{0} \, \forall j, k$$

$$Q \sim \chi^2_{(J-1)(k-1)}$$

Test H: No Effect of Age

Is the effect of `Age` across the three candidates equal to zero?

```
> library(aod)
> wald.test(b=c(t(coef(NES.MNL))),Sigma=vcov(NES.MNL),Terms=c(5,6))

Wald test:
----------

Chi-squared test:
X2 = 11.0, df = 2, P(> X2) = 0.0042
```

Test H: No Difference – Clinton vs. Bush

Are the estimated coefficients for Clinton (vs. Bush) jointly equal to zero?

```
> wald.test(b=c(t(coef(NES.MNL))),Sigma=vcov(NES.MNL),Terms=c(1,3,5,7,9))

Wald test:
----------

Chi-squared test:
X2 = 444.6, df = 5, P(> X2) = 0.0
```

Marginal efffects are:

$$\frac{\partial \text{Pr}(Y_i = j)}{\partial X_k} = \text{Pr}(Y_i = j|\mathbf{X}) \left[\hat{\boldsymbol{\beta}}_{jk} - \sum_{j=1}^{J} \hat{\boldsymbol{\beta}}_{jk} \times \text{Pr}(Y_i = j|\mathbf{X}) \right]$$

Depend on:

- $\text{Pr}\widehat{(Y_i = j)}$
- $\hat{\beta}_{jk}$
- $\sum_{j=1}^{J} \hat{\boldsymbol{\beta}}_{jk}$

Available for `-multinom-` (in the `-nnet-` package) via the `-margins-` package...

Marginal Effects: Illustrated

```
> MNL.alt<-multinom(PresVote2~PartyID+Age+White+Female,data=NES92,Hess=TRUE)
# weights:  18 (10 variable)
initial  value 1618.255901
iter  10 value 1080.908630
final  value 1053.650588
converged

> summary(marginal_effects(MNL.alt))
  dydx_PartyID          dydx_Age           dydx_Female        dydx_WhiteWhite
 Min.   :-0.0908   Min.   :-0.00362   Min.   :-0.1158   Min.   :0.0416
 1st Qu.:-0.0439   1st Qu.:-0.00282   1st Qu.:-0.0892   1st Qu.:0.0943
 Median : 0.0185   Median :-0.00215   Median :-0.0674   Median :0.1352
 Mean   : 0.0083   Mean   :-0.00207   Mean   :-0.0648   Mean   :0.1435
 3rd Qu.: 0.0618   3rd Qu.:-0.00138   3rd Qu.:-0.0420   3rd Qu.:0.1848
 Max.   : 0.1070   Max.   :-0.00011   Max.   :-0.0034   Max.   :0.2926
```

Odds ("Relative Risk") Ratios

MNL has:

$$\ln\left[\frac{\Pr(Y_i = j|\mathbf{X})}{\Pr(Y_i = j'|\mathbf{X})}\right] = \mathbf{X}(\hat{\boldsymbol{\beta}}_j - \hat{\boldsymbol{\beta}}_{j'})$$

Setting $\hat{\boldsymbol{\beta}}_{j'} = \mathbf{0}$:

$$\ln\left[\frac{\Pr(Y_i = j|\mathbf{X})}{\Pr(Y_i = j'|\mathbf{X})}\right] = \mathbf{X}\hat{\boldsymbol{\beta}}_j$$

One-Unit Change in X_k:

$$RRR_{jk} = \exp(\boldsymbol{\beta}_{jk})$$

δ-Unit Change in X_k:

$$RRR_{jk} = \exp(\boldsymbol{\beta}_{jk} \times \delta)$$

Odds ("Relative Risk") Ratios

```
> mnl.or <- function(model) {
    coeffs <- c(t(coef(NES.MNL)))
    lci <- exp(coeffs - 1.96 * diag(vcov(NES.MNL))^0.5)
    or <- exp(coeffs)
    uci <- exp(coeffs + 1.96* diag(vcov(NES.MNL))^0.5)
    lreg.or <- cbind(lci, or, uci)
    lreg.or
  }

> mnl.or(NES.MNL)
                   lci      or     uci
(Intercept):1 139.540 332.504 792.309
(Intercept):2   2.591   7.243  20.250
PartyID:1       0.288   0.321   0.358
PartyID:2       0.551   0.606   0.666
Age:1           0.987   0.997   1.008
Age:2           0.975   0.985   0.994
WhiteWhite:1    0.201   0.372   0.688
WhiteWhite:2    1.025   2.409   5.662
Female:1        0.634   0.882   1.229
Female:2        0.437   0.601   0.827
```

Odds Ratios: Interpretation

Odds ratio interpretations:

- A one unit increase in `partyid` corresponds to:
  - A decrease in the odds of a Clinton vote, versus a vote for Bush, of $\exp(-1.136) = 0.321$ (or about 68 percent), and
  - A decrease in the odds of a Perot vote, versus a vote for Bush, of $\exp(-0.501) = 0.606$ (or about 40 percent).
  - These are *large* decreases in the odds – not surprisingly, more Republican voters are *much* more likely to vote for Bush than for Perot or Clinton.
- Similarly, `female` voters are:
  - No more or less likely to vote for Clinton vs. Bush (OR=0.88), but
  - Roughly 40 percent less likely to have voted for Perot (OR=0.60).

$$
\begin{aligned}
\widehat{\Pr(\texttt{presvote}_i} = \text{Bush}) &= \frac{\exp(\mathbf{X}_i \hat{\boldsymbol{\beta}}_{\text{Bush}})}{\sum_{j=1}^{J} \exp(\mathbf{X}_i \hat{\boldsymbol{\beta}}_j)} \\
&= \frac{1}{1 + \sum_{j=2}^{J} \exp(\mathbf{X}_i \hat{\boldsymbol{\beta}}_j)}
\end{aligned}
$$

In-Sample Predicted Outcomes

Generate predicted vote choices:

```
> NES92$Predictions<-" "
> NES92$Predictions<-ifelse(fitted.values(NES.MNL)[,1]>fitted.values(NES.MNL)[,2]
+                  & fitted.values(NES.MNL)[,1]>fitted.values(NES.MNL)[,3],
+                  paste("Bush"),NES92$Predictions) # Bush
> NES92$Predictions<-ifelse(fitted.values(NES.MNL)[,2]>fitted.values(NES.MNL)[,1]
+                  & fitted.values(NES.MNL)[,2]>fitted.values(NES.MNL)[,3],
+                  paste("Clinton"),NES92$Predictions) # Clinton
> NES92$Predictions<-ifelse(fitted.values(NES.MNL)[,3]>fitted.values(NES.MNL)[,1]
+                  & fitted.values(NES.MNL)[,3]>fitted.values(NES.MNL)[,2],
+                  paste("Perot"),NES92$Predictions) # Perot)

> # "Confusion Table":
>
> table(NES92$VotedFor,NES92$Predictions)

        Bush Clinton Perot
  Bush     415      77     8
  Clinton   56     619    16
  Perot    135     133    14
```

$\rightarrow$ PRE

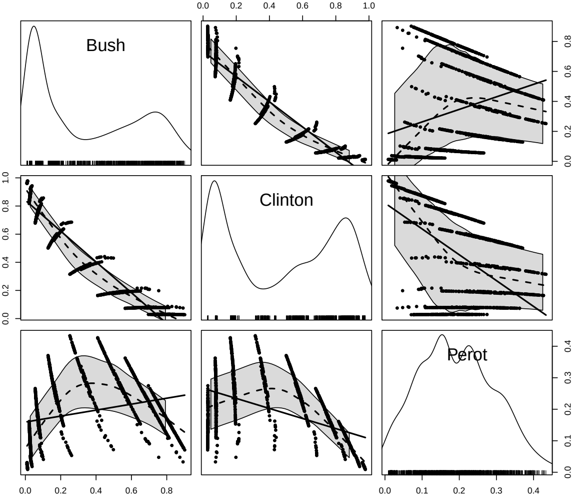
Model fit:

- "Null" Model: $\left(\frac{691}{1473}\right) = 46.9\%$ correct.
- Estimated model: $\frac{(415+619+14)}{1473} = \frac{1048}{1473} = 71.2\%$ correct.
- PRE $= \frac{1048-691}{1473-691} = \frac{357}{782} = 45.7\%$.
- Correct predictions: 90% Clinton, 83% Bush, 5% Perot.

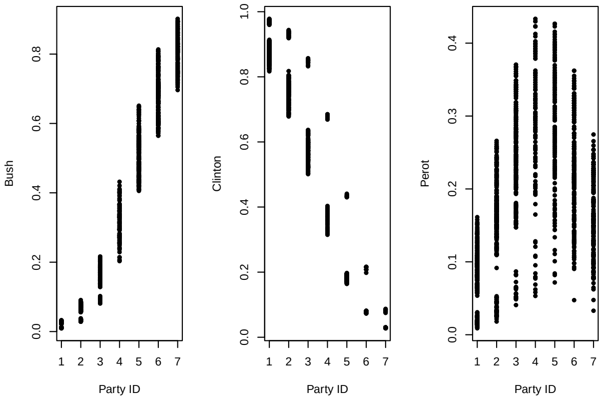
In-Sample Predicted Probabilities

```
> hats<-as.data.frame(fitted.values(NES.MNL))
> names(hats)[3]<-"Perot" # nice names...
> names(hats)[2]<-"Clinton"
> names(hats)[1]<-"Bush"
> attach(hats)

> library(car)
> scatterplot.matrix(~Bush+Clinton+Perot,
  diagonal="histogram",col=c("black","grey"))
```

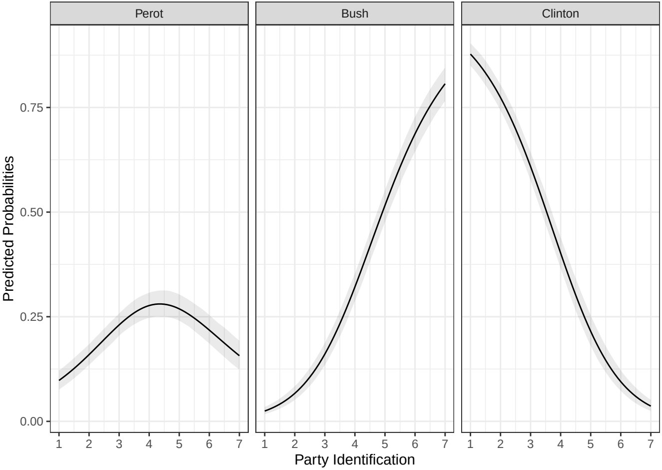
In-Sample $\widehat{\text{Prs}}$



In-Sample $\widehat{\text{P}}$rs vs. `partyid`

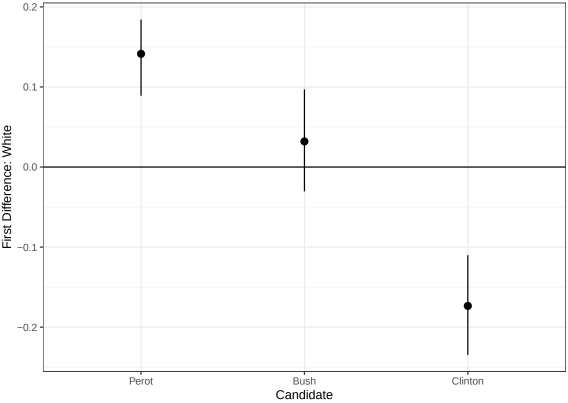


Out-Of-Sample Predictions (using `MNLpred`)



OOS First Differences (using `MNLpred`)

First differences in probabilities associated with `White`:

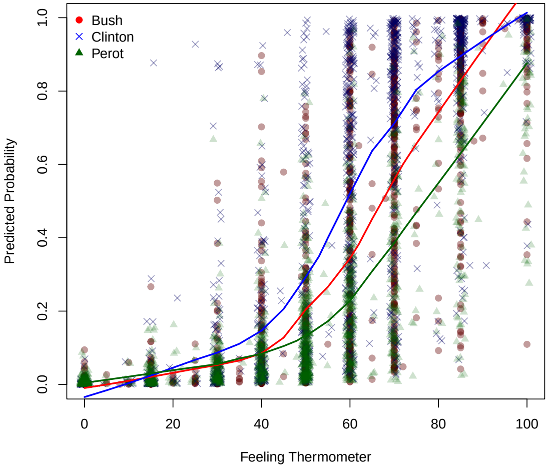


Conditional Logit: Example

```
> summary(NES92.clogit2)

Call:
mlogit(formula = VotedFor ~ FT | PartyID + Age + White + Female,
    data = AltNES92, reflevel = "Perot", method = "nr")

Frequencies of alternatives:choice
  Perot    Bush Clinton
  0.191   0.339   0.469

nr method
6 iterations, 0h:0m:0s
g'(-H)^-1g = 3.06E-08
gradient close to zero

Coefficients :
                    Estimate Std. Error z-value Pr(>|z|)
(Intercept):Bush    -0.39416    0.58730   -0.67  0.50214
(Intercept):Clinton  2.69235    0.50403    5.34  9.2e-08 ***
FT                   0.06231    0.00325   19.17  < 2e-16 ***
PartyID:Bush         0.22298    0.05842    3.82  0.00014 ***
PartyID:Clinton     -0.40620    0.05798   -7.01  2.4e-12 ***
Age:Bush             0.00868    0.00612    1.42  0.15639
Age:Clinton          0.00839    0.00598    1.40  0.16040
WhiteWhite:Bush     -1.31961    0.47725   -2.77  0.00569 **
WhiteWhite:Clinton  -1.57156    0.41404   -3.80  0.00015 ***
Female:Bush          0.39271    0.19995    1.96  0.04953 *
Female:Clinton       0.28585    0.19474    1.47  0.14213
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -723
McFadden R^2:  0.527
Likelihood ratio test : chisq = 1610 (p.value = <2e-16)
```

Conditional Logit: In-Sample Predicted Probabilities

```
> CLhats<-predict(NES92.clogit2,AltNES92)
```



Various things:

- "Independence of Irrelevant Alternatives"
- $\rightarrow$ Multinomial Probit
- $\rightarrow$ Heteroscedastic Extreme Value model
- "Mixed" Logit
- Nested Logit, etc.